



Certificate of Achievement

Jordan KAMTA NLONFO

has completed the following course:

FUNDAMENTALS OF OCEAN FORECASTING

ICMPA - UNESCO CHAIR, GULF OF GUINEA OCEAN SCIENCES SUMMER SCHOOL (GGOSSS),
UNIVERSITÉ D'ABOMEY-CALAVI AND OCEAN PREDICTION DCC

Explored the end-to-end ocean forecasting system - from ocean dynamics and numerical models to observations, data assimilation, upstream data sources, forecast interpretation, and downstream applications supporting maritime safety, coastal management, and sustainable ocean decision-making.

7 weeks, 4 hours per week



Prof. Ezinvi BALOÏTCHA

CHAIR Holder, International Chair of Mathematical
Physics and Applications (ICMPA-UNESCO CHAIR)
ICMPA - UNESCO Chair



Dr. Babette C. Tchouang

Assistant Research Scientist, Florida State University
Gulf of Guinea Ocean Sciences Summer School
(GGOSSS)



The person named on this certificate has completed the activities in the attached transcript. For more information about Certificates of Achievement and the effort required to become eligible, visit futurelearn.com/proof-of-learning/certificate-of-achievement.

This certificate represents proof of learning. It is not a formal qualification, degree, or part of a degree.



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96%
OVERALL
SCORE

Successfully completed OPERA MOOC Set 1: Fundamentals of Ocean Forecasting – a seven-week course developed by the ICMPA–UNESCO Chair and GGOSSS, under the EU-funded OPERA project implemented by Mercator Ocean International. This course received funding from the European Union through the Ocean Prediction Enhancement of Regions in Africa (OPERA project) in the framework of the African Regional Centres of Excellence for Science, Technology and Innovation for the green transition (ArcX).

STUDY REQUIREMENT

7 weeks, 4 hours per week

LEARNING OUTCOMES

- Describe the architecture and purpose of an end-to-end ocean forecasting system, from observations to user applications
- Explain the fundamental physical principles governing ocean circulation, including geostrophic balance, Ekman dynamics, tides, waves, and mesoscale variability
- Explain how numerical ocean models are constructed, configured, and used in operational forecasting at global, regional, and coastal scales
- Identify the major satellite and in-situ observing systems that supply ocean forecasting systems, and explain the role of data assimilation in improving forecast accuracy
- Identify and access key upstream data sources including Copernicus Marine Service products and atmospheric forcing datasets

- Interpret common ocean forecast outputs such as sea surface temperature maps, current fields, and sea level anomalies, and recognize their limitations
- Explain how ocean forecast products are translated into downstream applications across sectors such as fisheries, maritime safety, and coastal risk management, with examples from Africa

SYLLABUS

- Week 1 – Overview of the end-to-end ocean forecasting system
- Week 2 – Fundamentals of ocean dynamics
- Week 3 – Introduction to numerical ocean models
- Week 4 – Ocean observations and data assimilation
- Week 5 – Upstream data sources for ocean forecasting
- Week 6 – Interpretation of ocean forecast outputs
- Week 7 – Introduction to downstream applications of ocean forecasts